Lab: Blender Animation

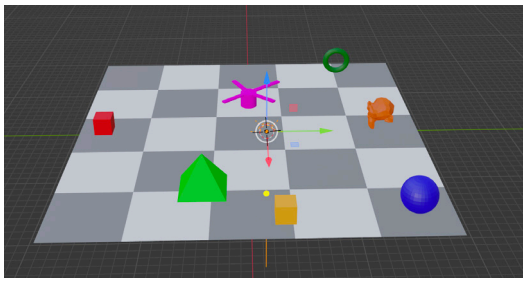


Finally after weeks and months of learning about 3D modeling, materials, lighting, rigging and more we are finally able to get to the good stuff, which is animating in Blender!

In this lab we will cover the basics of the animating in Blender with the timeline as well as creating keyframes.

Overview

Make sure you have download *Lab-Blender-Animation-Basics* from the Canvas. Open *Lab-Animation.blend* into Blender.

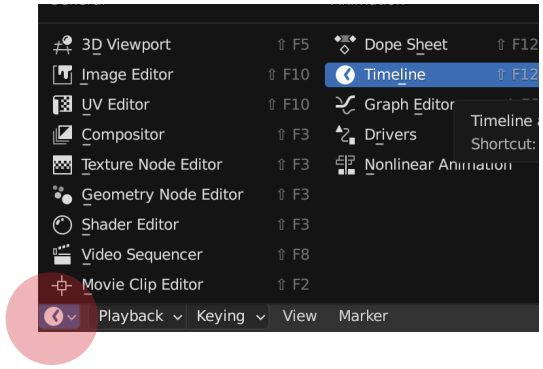


Click below to view the animation we are going to create:

https://youtu.be/uzwMqF0yo8U

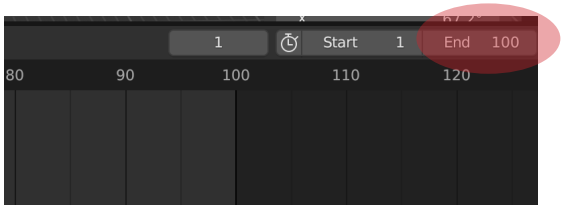
You can see that there is a checkboard and various 3D models are moving around. When you open the Blender file you will see all of the objects in the world, we just need to animate them!

Timeline Basics

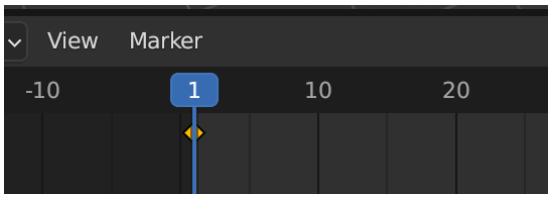


To animate in Blender we will need the Blender *Timeline*. Typically the *Timeline* can be found at the bottom of the user interface. If you do not have one you can create a new user-interface window and select *Timeline* from the *Editor Type* option.

You can navigate through the timeline like you would with the scene. To zoom in / zoom out of the timeline you can use the scroll wheel of your mouse. To pan through the timeline you can click on hold the scroll wheel button.



The timeline, as it sounds, shows the progression of time in your project. In Blender time is measured in frames. A frame is single photo or render. In other words, a frame is fraction of a second. The number on the far right is the End frame, or how many frames total there are in the animation. This project the timeline is a total of 100 frames.



To know where in time you are you need to view the Time Marker. The Time Marker is the blue vertical line in the Timeline. The number at the top is the current frame of the animation.

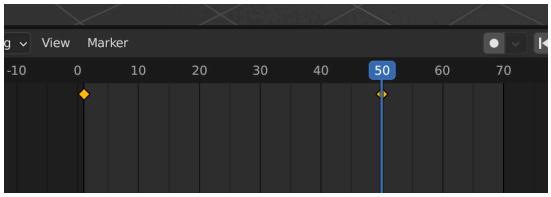
Keyframe Basics

A keyframe is a saved point in time for an object in our project. In Blender object / characters will interpolate or move between saved points for location, rotation, scale and other properties over time.

Select the *Gold Cube* in the scene by selecting it from the *3D*

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Viewport or from the *Outliner*. You will see in the *Timeline* that two diamonds appear, those are keyframes. The two keyframes are saved points of information, in this case the starting location of the *Gold Cube* and ending location of the *Gold Cube*.

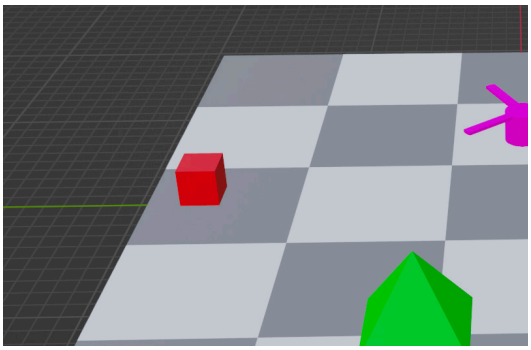


Deselect the *Gold Cube* by clicking off of it. You will notice that the diamonds disappear. You can only view keyframes in the Timeline for an object when it is highlighted.

Animate with Keyframes

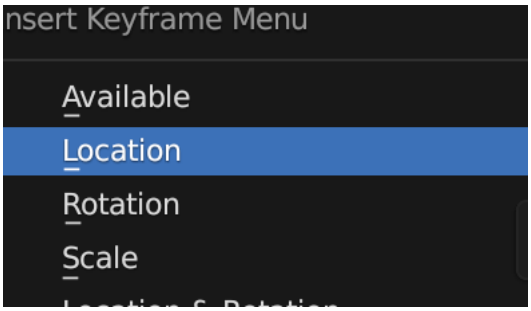
Now that we know what keyframes are, let's create our own to animate the *Red Cube*. We want to animate the *Red Cube* to move from the edge of the board to the center of the board.

When animating we need to know two things, what is the action that is occuring and when is it occuring. The *'action'* is whether the object is moving, scaling, rotating, etc. The *'when'* is when the start of that action is and when the end of the action is. If you view the animation link you can see that the *Red Cube's* action is moving, and it begins at the start of the animation or frame 1 and goes for two seconds.

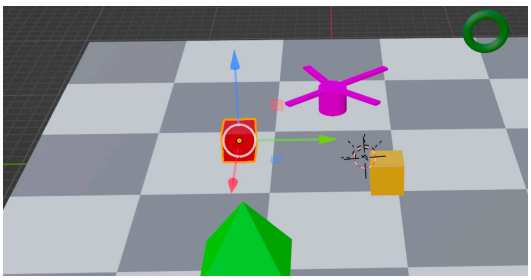


Let's begin to animate this. Start by setting the time marker in the timeline to the beginning or *frame 1*. Select the *Red Cube* and use the *Move tool* to set where the object should start. As stated, it should be on the edges of the checkerboard.

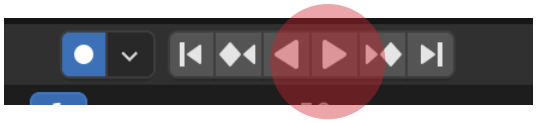
To set a keyframe or save this location in time press the **'i'** key on your keyboard **'i'** stands for *insert*. This will create a pop up menu to appear with the various properties we can keyframe, since we are moving the *Red Cube* we want to keyframe the *'Location'* property, so choose *Location* from the popup list. After doing this you should see a diamond appear in the *Timeline*, that is your *location* keyframe.



Now that we set the beginning of the action, we need to set the end of the action. We want the *Red Cube* to move for two seconds so let's move the time marker to *frame 50* in the timeline. Use the *Move Tool* to move the *Red Cube* to the center of the checker board, location *X = 0, Y=-6m, Z =1.5m*. Once again use the **'i'** key to open the pop up menu and choose *'Location',* this will create a second keyframe. Your timeline should now have two keyframes in it: one at *frame 1* and one of *frame 50*.



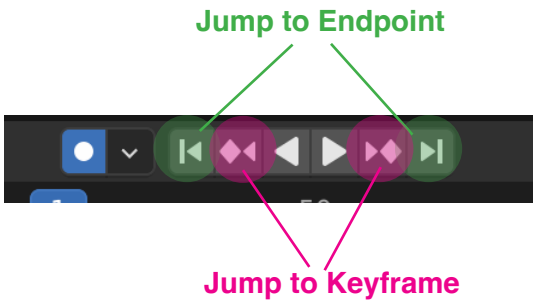
Preview your animation by dragging the time marker back to the start of the timeline and pressing the *Play* button. This play button looks like the play button on common video players that you would see in Hulu and Netflix. The result is your animation you should play and the *Red Cube* will move across the checkerboard.



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Playback Controls

The timeline has playback controls to assist with playing your animation and jumping to different parts of the timeline. As mentioned the *Play* button will play all of frames. There is also a reverse playback button that will play your timeline in reverse.

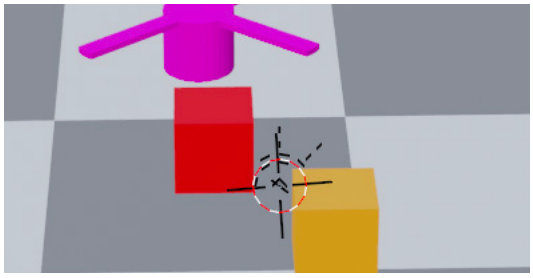


There is also a *Jump to Endpoint (beginning)* it will always return to the first frame in the timeline. The *Jump to Endpoint (end)* will also go to last frame in the timeline, in this case that is *frame 100*.

Finally, there is *Jump to Keyframe (previous)* and *Jump to Keyframe (next)*. This allows you to move quickly just between keyframes in the project.

Edit a Keyfame

You can also edit the value of a keyframe, even after it has been set. For our *Red Cube*, we would actually like the cube to come closer to the center of the board. To do this we will reset the value for this keyframe.

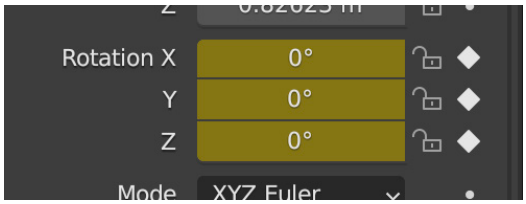


Use the *Jump to Keyframe* button in the timeline to land on the second keyframe. In the *Object Properties* change location to *X: 0, Y:0, Z: 1.5m*. Press **"i"** to set the keyframe for this and viola, the keyframe has been updated.

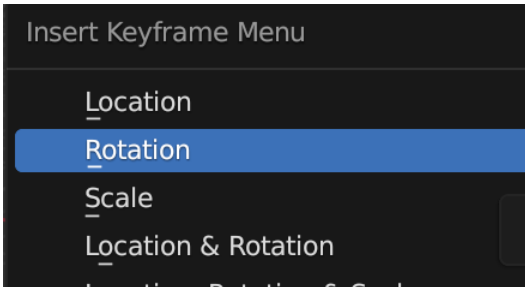
Scrub through the timeline and you will see that the *Red Cube* now comes in farther.

Rotation Animation

In addition to moving or changing the location of an object in our scene we can also rotate an object as well. If you view the preview animation you can see that the *Purple Fan* in the scene spins or rotates, let's see if we can do this with keyframes.



Move the time marker back to frame 1 in the timeline. Select the *Purple Fan* in the scene. Use the *Rotate* tool to set at what angle the fan should start at, in the example the fan starting rotation for the *z-axis is 0*.

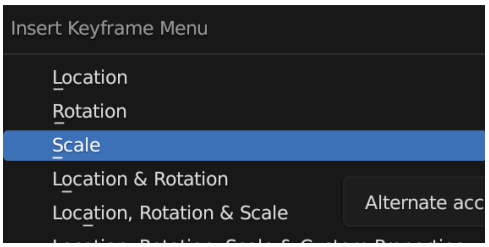


Set a keyframe by pressing the **'i'** key on your keyboard. Choose *"Rotation"* from the pop up menu. A new keyframe should appear. Move the time marker to *frame 50.* Use the *Rotate* tool to rotate the fan to *720˚* for the z axis. or two full rotations.

Save and scrub through the timeline and the *Purple Fan* should now spin.

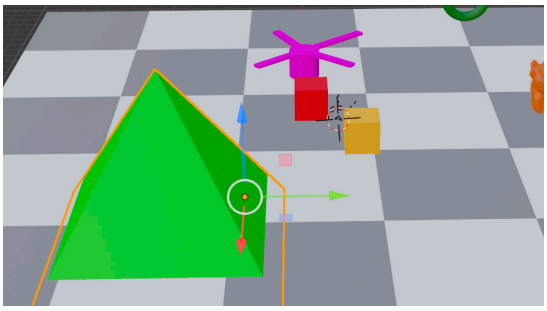
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Scale Animation



In addition to moving and rotating we can also scale items as well. If you view the preview animation you can see that the *Green Pyramid* in the scene grows or scales over time.

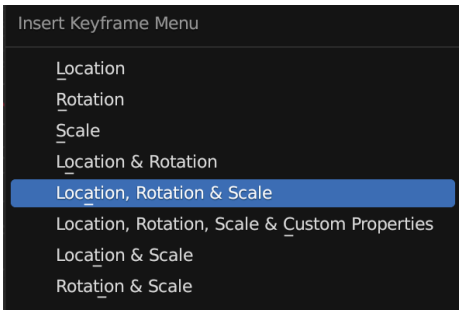
Move the time marker back to frame 1 in the timeline. Select the *Green Pyramid* in the scene. Use the *Scale* tool to set the size at which the pyramid will start. In the example the scale size is x/y/z being 1.0.



Set a keyframe by pressing the **'i'** key on your keyboard. Choose *"Scale"* from the pop up menu. A new keyframe should appear. Move the time marker to *frame 50.* Use the *Scale* tool to scale the *Green Pyrmaid* to 2.5 in the x,y and z axis.

Scrub through the timeline and the *Green Pyramid* should grow.

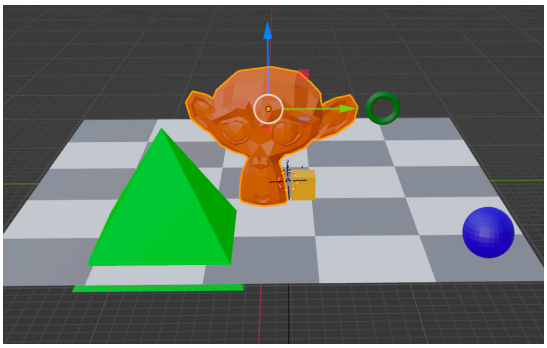
Location / Rotation / Scale



In the last examples we animated just one property such as location or rotation, but if we want we actually animate all three at once.

If you view the preview animation you'll see that the *Orange Monkey* moves, rotates and scales. Let's try to do the same.

Begin by moving the time marker back to frame 1 in the timeline. Highlight the *Orange Monkey* and use the Move / Rotate and Scale tool to set a starting location, rotation and scael for the object. Press **"i"** and choose *"Location, Rotation and Scale"* from the pop up menu.



Move the the time marker to *frame 50*. Use the Move, Rotate and Scale tool to manipulate the cube. Feel free to add your own values. When finished press **"i"** and choose *"Location, Rotation and Scale"* from the pop up menu to set your end keyframe.

Save and scrub through the timeline and watch as the Orange Monkey animates through all of the properties.

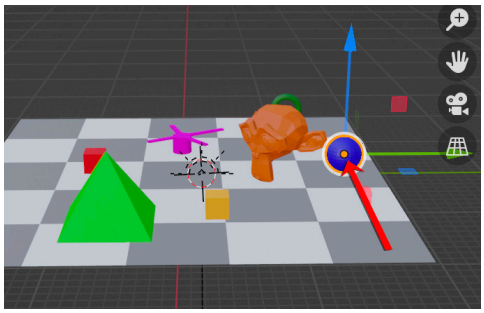
Multiple Keyframes

In the past examples we have done simple animations that have an action with a start and an end, or an A → B action. Sometimes though we may want to animate a more complex animation where the character moves from A → B, B → C and C → D, etc. This is actually easy to do in Blender.

If you view the preview animation you'll see that the *Blue Sphere* moves along the x-axis, then moves along the y-axis and then finally moves along the z-axis. Let's try to do the same.

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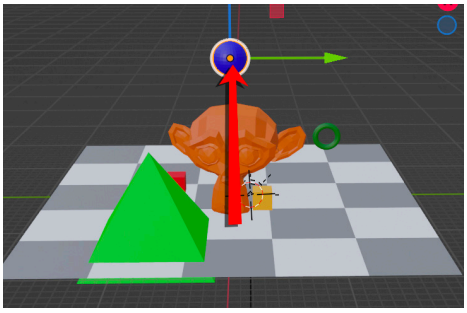
Begin by moving the time marker back to frame 1 in the timeline. Highlight the *Blue Sphere* and set the location to *X: 16, Y: 16, Z: 2*. Set a *Location* keyframe at this point.



Move to *frame 25* and set the *Location x-value* of the *Blue Sphere* from *16* to *0*. Set a Location keyframe for this point.

Move to *frame 50* and and set the *Location y-value* from *16* to *0*. Set a *Location* keyframe at this point.

Move to *frame 75* and set the *Location z-value* from *2* to *15*. Set a *Location* keyframe at this point.

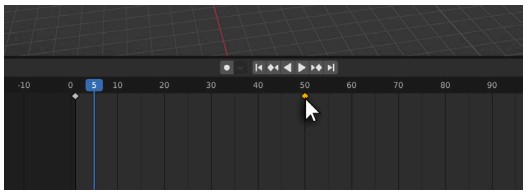


Finally move to *frame 100* and set the *Location z-value* from *15* to *0*. Set a *Location* keyframe at this point.

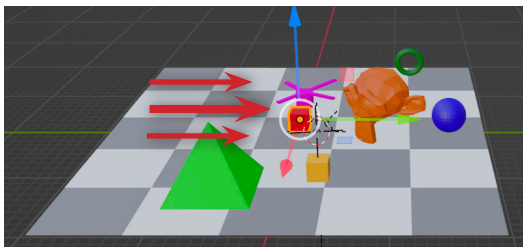
Save and scrub through your animation. You should see that the Blue Sphere goes from point A → B, then from Point B → C ,from Point C → Point D and finally from Point D → E. This process allows you to add multiple keyframes for an object in a project.

Move Keyframes

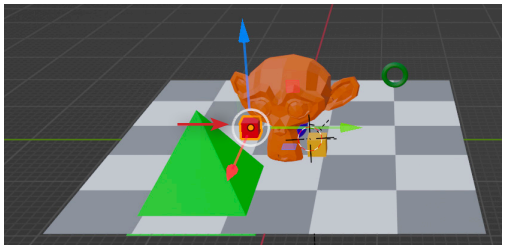
After setting keyframes you may want to move in the Timeline. One reason for moving a keyframe is to change when an action starts or ends, but also to change the speed of an action as well.



Play back the animation and watch the *Red Cube* move across the board. We may decide to speed up or slow down this action. In animation when two keyframes are brought closer in the timeline the action between the two becomes quicker, when two subsequent keyframes are brought farther from one another the action will go slower.



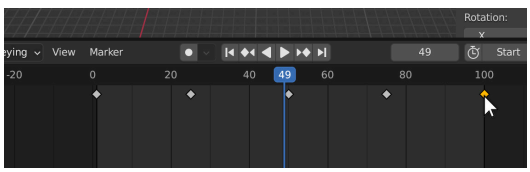
Select the *Red Cube* to bring the keyframes up in the *Timeline*. Let's take the second keyframe and move it from *frame 50* to *frame 25*. Use the mouse to select the keyframe and drag it to the right to *frame 25*. Save and play the animation, you will see that the Red Cube is moving faster than before.



Select the *Red Cub*e again. Select the second keyframe on *frame 25* and move it f*rame 75*. Save and play the animation and you will see that the *Red Cube* moves slower now.

Delete Keyframes

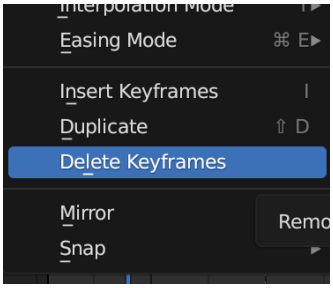
There will be times when animating instead of editing a keyframe, we simply want to remove it. This is typically when we want to remove all action from a character or perhaps remove an unwanted keyframe as part of a larger animation.



Looking back on the *Blue Sphere*, instead of the ball returning to

the board at the end we want it to stay in the air. To do this, we can delete the last keyframe for that object.

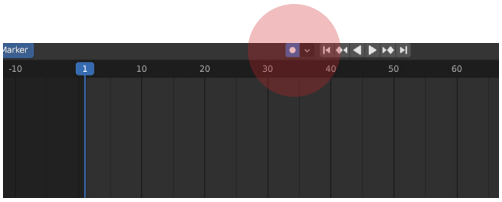
Select the *Blue Sphere* to view the timeline. Use the mouse to select the final keyframe on *frame 100.* Right click and from the pop up menu and choose *Delete Keyframes* or go to *Object > Animation > Delete Keyframes*.



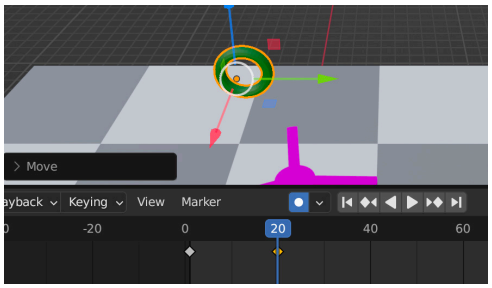
The result will be the final keyframe will be deleted. Save and scrub through the animation and you will see that the *Blue Sphere* animation stops after it moves upward and does not come back down.

Auto Keying

Sometimes during the process of animating instead of needing to press **"i"**, it would nice that whenever we change a value that Blender would automatically create a keyframe for us. Well, we can do this, it is referred to as *Auto Keying*.

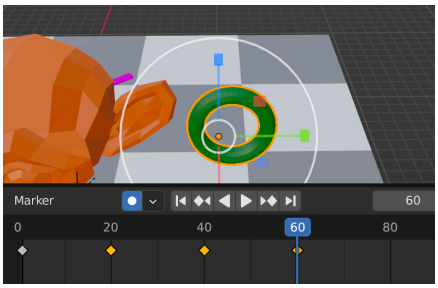


The *Auto Keying* button in on the left side of the timeline. Once it is turned on anytime you move, rotate, scale or change a property on the object it will automatically be keyframed for you. Let's apply this to the *Green Donut*.



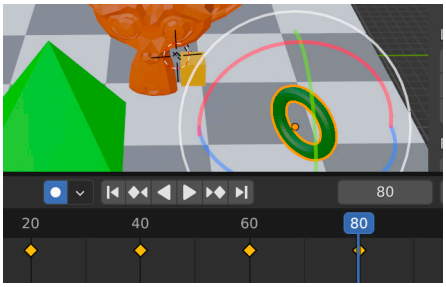
Begin by moving the time marker back to *frame 1* in the timeline. . Select the *Green Donut* in the scene and set a *Location* keyframe on *frame 1*. Then press the Auto Keying button on the timeline.

Move the time marker to *frame 20* and use the *Move Tool* to move the *Green Donut* to a new location. When you do this you'll notice Blender automatically makes a new keyframe.



Move the time marker to *frame 40* and once again move the the *Green Donut* to a new location. Move the time marker to *frame 60* and move the *Green Donut* once again. In addition to moving the donut use the *Scale Tool* to resize the *Green Donut* as well.

Finally, move the time marker to *frame 80*. Use the *Move Tool* to move the *Green Donut* to a new location. Use the *Rotate Tool* to rotate the *Green Donut* as well.



Hit the *Auto Keying* button again to turn off auto keying. Scrub through your animation and you'll see that auto keying helped to animate the move, scale and rotate of the character.

Submission

Whew! That was a lot, but hopefully you have a basic idea of how to perform basic animation and apply keyframes in Blender.

Submit *Lab-Animation-Blender-Basics* to Canvas for grading.

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